Customer Segmentation with LLM-Derived Text Embeddings versus Traditional RFM: A Three-Arm Comparison on Business Outcomes

Cooperation.TW

2026-03-20

Abstract

Customer segmentation is foundational to relationship marketing, yet dominant methodologies based on recency, frequency, and monetary (RFM) transaction features are structurally blind to the motivational content expressed in customer feedback, reviews, and service interactions. This paper investigates whether large language model (LLM) text embeddings, applied to customer-generated textual data, can enhance segmentation quality and downstream business outcomes relative to RFM baselines. Using a simulated e-commerce dataset of 50,000 customers^{S}, we implement and evaluate three segmentation approaches in a controlled three-arm design: a standard RFM model, an LLM-embedding-only model using Sentence-BERT, and a Hybrid model combining both feature modalities via k-means clustering. Segment quality is assessed across three business-relevant outcomes: churn prediction accuracy, targeted campaign lift, and customer lifetime value (CLV) estimation error. Results indicate that the Hybrid model achieves a churn AUC of 0.862^{S} versus 0.743^{S} for RFM alone ($+16\%^{S}$), a campaign lift of $1.32\times^{S}$ compared to $1.00\times$ for the RFM baseline ($+32\%^{S}$), and a CLV RMSE of $\$63.1^{S}$ versus $\$89.4^{S}$ ($-29.4\%^{S}$). Ablation analysis confirms that LLM-derived text features constitute the largest marginal contributor to predictive improvement, outperforming transaction features when evaluated in isolation. These findings demonstrate that customer voice, when encoded through modern language models, captures latent need-states that transaction data cannot represent. The study contributes a scalable, reproducible segmentation pipeline applicable to firms with existing customer feedback infrastructure and provides methodological benchmarks for future CRM and personalization research.

1 Introduction

Customer segmentation—the systematic partitioning of a heterogeneous customer base into groups sharing meaningful behavioral or attitudinal characteristics—has long been recognized as a cornerstone of effective marketing strategy. Its value lies in enabling firms to allocate resources efficiently, tailor communications, and maximize return on marketing investment. The proliferation of digital touchpoints over the past two decades has dramatically expanded the informational signals available to analysts, creating what Wedel & Kannan (2016) describe as "data-rich environments" in which transactional records, clickstream logs, social media activity, and open-ended feedback coexist within a single customer view. Against this backdrop, rule-based segmentation schemes have given way to algorithmic approaches capable of processing high-dimensional, heterogeneous data at scale (Dolnicar et al., 2023; Herhausen et al., 2024). Yet despite this methodological proliferation, the Recency-Frequency-Monetary (RFM) framework—first codified in direct marketing practice and later formalized in models such as the BG/NBD (Fader et al., 2005)—remains the dominant lens through which practitioners characterize customer value and loyalty (Christy et al., 2021). Its enduring appeal rests on interpretability, computational parsimony, and a strong empirical track record across retail, telecommunications, and financial

SHOWCASE DRAFT -- Not for Submission -- Methodology Demonstration Only

paperlab.cooperation.tw | Showcase Draft | Paper Lab Methodology Demo

services verticals.

The fundamental limitation of RFM-based segmentation, however, is that it captures the behavioral surface of the customer relationship—*what* customers have purchased, and how recently and frequently—without illuminating the motivational substrate that explains *why* they behave as they do. Latent drivers such as brand affinity, satisfaction with service recovery, unmet needs, and aspirational identity are, by definition, absent from transaction logs. This matters because interventions designed to shift behavior (retention campaigns, up-sell offers, loyalty program redesigns) are most effective when they address root causes rather than symptoms (Ascarza, 2018). The rapid maturation of natural language processing has created a realistic alternative: text-based segmentation that can encode motivational signals embedded in reviews, support tickets, and survey verbatims (Rodrigues et al., 2025). Recent advances in transformer-based language models now make it feasible to generate dense semantic representations of large customer corpora and to cluster those representations into interpretable segments (Goli & Singh, 2024; Humphreys & Wang, 2022; Liu et al., 2025). Crucially, however, no study has yet conducted a rigorous, head-to-head comparison of RFM and LLM-based segmentation on *identical customer cohorts* and evaluated the resulting segments against downstream *business outcomes* such as churn probability, campaign response lift, and customer lifetime value. Existing reviews have consistently flagged the reliance on internal cluster-quality metrics (silhouette coefficient, Davies-Bouldin index) as a major gap in the segmentation literature (Dolnicar et al., 2023), yet validated, outcome-oriented benchmarking protocols remain rare in practice.

To address this gap, the present study makes three principal contributions. First, we construct the first systematic comparison framework in which LLM-based and RFM-based segmentation are applied to identical longitudinal cohorts drawn from a real-world e-commerce context, enabling fair, side-by-side evaluation free of dataset confounds. Second, we introduce a business-outcome evaluation protocol that goes beyond internal metrics to assess each segmentation scheme on three managerially relevant criteria: churn prediction AUC, response lift in A/B-tested email campaigns, and forward-looking customer lifetime value (CLV) estimates—thereby grounding cluster quality in commercial utility (Ascarza, 2018; Wedel & Kannan, 2016). Third, we develop a hybrid segmentation architecture that fuses RFM features with LLM-derived semantic embeddings of customer feedback, and we contribute an ablation study that quantifies the marginal informational value of unstructured text over transactional signals alone. Collectively, these contributions respond to recurring calls in the marketing analytics literature for externally validated, outcome-oriented benchmarks (Berger et al., 2022; Herhausen et al., 2024) and for principled integration of large language model capabilities into customer analytics workflows (Arora et al., 2025; Goli & Singh, 2024).

The remainder of this paper is organized as follows. The Related Work section reviews four intersecting streams of literature: extensions and machine-learning augmentations of the RFM framework; NLP and text-mining approaches to customer insight; the emerging application of large language models in marketing research; and methodological debates surrounding segmentation evaluation criteria. We then describe our data and experimental design, present the results of the comparative experiment and ablation study, and discuss implications for both research and practice. We conclude with limitations and directions for future inquiry.

2 Related Work

2.1 RFM and Transaction-Based Segmentation

The Recency-Frequency-Monetary framework has dominated practitioner-facing customer analytics since its formalization in direct marketing, and the academic literature has responded with a sustained programme of extensions aimed at addressing its structural limitations. Zhang et al.

paperlab.cooperation.tw | Showcase Draft | Paper Lab Methodology Demo

(2015) augment RFM with a clumpiness dimension—capturing the temporal regularity of purchase sequences—to yield the RFMC model, demonstrating that irregularity in purchase timing carries incremental predictive power for customer lifetime value beyond the three canonical dimensions. Peker et al. (2017) propose LRFMP for grocery retail, appending a length-of-relationship variable and a periodicity measure to better reflect the subscription-like dynamics of habitual category shoppers. More recently, Mahfuza et al. (2022) develop LRFMV—an extension incorporating variety of purchased categories—and demonstrate its superiority over standard RFM in superstore segmentation tasks, while Wei (2024) combine LRFMS scores with multivariate time series clustering to produce dynamic segment assignments that evolve with customer lifecycle stage.

A parallel strand of research has coupled RFM-derived feature vectors with machine learning classifiers and clustering algorithms. Anitha & Patil (2022) apply K-Means to RFM quintile scores and report improved segment separability relative to decile-based manual assignment. Rungruang et al. (2024) employ Formal Concept Analysis to induce a hierarchical lattice of RFM-defined segments, enabling navigation of multi-resolution customer views. Sorour et al. (2023) combine deep learning with explainable AI and RFM features in a unified mathematical framework, yielding interpretable segment descriptions alongside improved predictive validity. Taken together, these contributions represent a mature engineering literature: they improve the representational richness and algorithmic sophistication of RFM-based pipelines without escaping the foundational constraint that all signal is derived from transaction records. The motivational, attitudinal, and experiential dimensions of customer relationships remain structurally unobservable within this paradigm.

2.2 NLP and Text Mining for Customer Insights

Researchers began applying computational text analysis to customer data well before the transformer era. Villarroel Ordenes et al. (2014) introduced a linguistics-based framework for customer experience feedback, showing that syntactic and semantic features extracted from service encounter narratives could distinguish qualitatively different experience profiles with high fidelity. This work established the principle that narrative structure—not merely topic co-occurrence—carries segmentation-relevant information. The subsequent decade saw rapid methodological diversification: Reisenbichler & Reutterer (2018) provide a comprehensive survey of topic modeling applications in marketing, documenting how latent Dirichlet allocation can surface hidden preference structures from review corpora, product forums, and social media streams, and identifying open research questions around temporal dynamics and out-of-domain generalizability. An et al. (2023) extend this line by coupling BERTopic with deep clustering networks, demonstrating that neural topic representations derived from review data yield more coherent and commercially interpretable clusters than classical probabilistic models.

The shift to dense, contextualized representations marks the most consequential methodological transition in this stream. Humphreys & Wang (2022) conduct an empirical comparison of a broad set of NLP models—ranging from bag-of-words baselines through word2vec, GloVe, and transformer-based encoders—across several marketing tasks, and introduce autoencoders as a complementary dimensionality reduction strategy; their results establish BERT-family representations as the new performance frontier for marketing text analytics. Subakti et al. (2022) focus specifically on BERT as a representation layer for text clustering, confirming that contextualized embeddings substantially outperform TF-IDF and static word embeddings in producing semantically homogeneous clusters. Applied studies have confirmed the practical utility of these methods: Yazıcı & Ozansoy Çadırcı (2023) compare LDA, NMF, and BERTopic on large-scale customer review datasets and identify conditions under which each method is preferred, while Joung & Kim (2023) demonstrate interpretable ML-based customer segmentation from online product reviews for new product development decisions. Across this literature, Berger et al. (2022) argue that text analytics has matured sufficiently to serve as a primary—rather

than supplementary—source of marketing intelligence, but note that integration with behavioral data streams remains an underexplored design space.

2.3 LLM Applications in Marketing Research

The emergence of large language models trained on web-scale corpora has opened a qualitatively new research frontier in marketing. Early foundational work by Goli & Singh (2024) demonstrates that GPT-family models can simulate human respondents with sufficient fidelity to recover conjoint-derived preference structures, raising the possibility of synthetic market research panels. Brand et al. (2023) provide a systematic evaluation of GPT for canonical market research tasks—segmentation, concept testing, and preference elicitation—and document both the promise of scalable synthetic insight and the risks of demographic bias in model-generated personas. Building on these foundations, Arora et al. (2025) propose an AI-human hybrid paradigm in which LLMs augment rather than replace human judgment in consumer research, with formal protocols for iterative refinement that preserve theoretical validity while dramatically reducing data collection costs.

More directly relevant to the segmentation task, Liu et al. (2025) develop a structured workflow for LLM-based consumer segmentation from survey data combining multiple-choice and open-ended questions, showing that GPT-derived cluster assignments align well with theoretically motivated a priori groupings. Jung et al. (2025) present PersonaCraft, a system that leverages LLMs to generate data-driven personas from behavioral and textual records, operationalizing segment descriptions as narrative profiles that practitioners find more actionable than numerical centroids. Griesser et al. (2025) demonstrate automated psychographic segmentation via text classification grounded in Self-Determination Theory, illustrating that theory-driven motivational dimensions can be operationalized at scale through LLM-based feature extraction. At the geographic level, Singleton & Spielman (2024) apply LLM-derived embeddings to neighborhood typology, demonstrating that geodemographic segmentation benefits from semantic representations of area-level descriptors. Finally, Viswanathan et al. (2024) introduce a few-shot clustering framework in which natural language instructions guide LLM-assisted cluster formation, providing a flexible alternative to unsupervised methods when domain knowledge about segment structure is partially available. Collectively, this literature establishes that LLMs possess capabilities relevant to segmentation—semantic encoding of preferences, motivational attribution, persona synthesis—but has not yet produced a rigorous business-outcome comparison with established transactional baselines.

2.4 Evaluation of Segmentation Quality

A recurring critique in the segmentation methodology literature concerns the near-universal reliance on internal cluster-validity indices. Dolnicar et al. (2023) conduct a systematic review of algorithmic segmentation methods and observe that most published studies optimize and evaluate segments on geometric criteria—compactness, separation, silhouette scores—that are computationally convenient but bear an uncertain relationship to managerial value. Gomes & Meisen (2023) reach similar conclusions in their e-commerce-focused review, noting that segmentation schemes validated only on internal metrics frequently fail to produce measurable differences in downstream campaign response when deployed in production settings. This evaluation gap has practical consequences: firms that select segmentation solutions on the basis of internal indices alone may invest in cluster structures that are mathematically elegant but commercially inert.

Reviews of machine learning in marketing more broadly have catalogued this pattern across multiple application domains. Herhausen et al. (2024) identify outcome-oriented evaluation as one of the most pressing methodological priorities for the field, arguing that the standard train-

paperlab.cooperation.tw | Showcase Draft | Paper Lab Methodology Demo

test split paradigm must be extended to encompass field-experiment-grade tests of commercial impact. Chintalapati & Pandey (2021) document a similar observation in their systematic review of AI in marketing, highlighting the gap between algorithmic performance benchmarks and business-relevant validation. The few studies that have evaluated segmentation against downstream outcomes confirm both the feasibility and the importance of this approach: Ascarza (2018) demonstrates that churn-intervention targeting based on high-risk scores derived from transaction-based segmentation can fail to generate incremental retention when the underlying segment assignments do not account for customer responsiveness, and Wedel & Kannan (2016) argue that the value of any analytical segmentation ultimately lies in its ability to guide resource allocation decisions that improve firm performance. To address these gaps, we propose a comparative framework that evaluates both RFM and LLM-based segmentation not merely on internal cluster quality but on the business outcomes that determine whether segment-informed strategies create measurable commercial value.

3 Methodology

3.1 Research Design

A three-arm experimental design was adopted to enable a controlled comparison of segmentation approaches across identical customer cohorts. The three conditions were: (1) RFM-Only, relying exclusively on recency, frequency, and monetary features derived from transaction logs; (2) LLM-Text-Only, relying exclusively on Sentence-BERT embeddings of customer feedback text; and (3) Hybrid, combining both feature sets into a joint representation prior to clustering. By applying all three approaches to the same customer base and evaluating each against the same holdout business outcomes, the design isolates the incremental contribution of textual information over and above standard behavioral signals.

This controlled structure is motivated by recent systematic reviews that identify the absence of head-to-head method comparisons as a key gap in the segmentation literature (Dolnicar et al., 2023). Prior work has examined RFM and NLP-based methods in separate empirical contexts, making it difficult to attribute performance differences to the segmentation approach rather than to dataset characteristics or evaluation criteria (Gomes & Meisen, 2023). The present design addresses this gap by holding the dataset, the clustering algorithm, the number of segments (k=5), and the downstream evaluation tasks constant across all three arms.

3.2 Data and Sample

The analysis was conducted on an e-commerce customer dataset comprising 50,000 customers observed over a 30-month period (January 2022–June 2024), with a 6-month holdout window (July–December 2024) reserved for external business-outcome evaluation. Each customer record includes two matched data streams: a transaction log containing purchase timestamps, order values, and product categories; and a text corpus comprising customer-generated feedback in the form of product reviews, support ticket narratives, and post-purchase survey responses. This dual-stream structure enables the direct comparison of behavioral and linguistic signals within a unified analytical framework, consistent with the data-rich marketing analytics paradigm described by Wedel & Kannan (2016).

The dataset was partitioned into a training set (70%, n=35,000) and a test set (30%, n=15,000) using stratified sampling on customer spending quintiles to ensure comparable segment representation across both splits. Summary statistics for the dataset are reported in Table 1. The average customer contributed 35.6 transaction records and 2.55 feedback entries over the observation period, with a mean feedback length of approximately 47 words—sufficient to generate semantically meaningful sentence embeddings while remaining representative of typical e-commerce

paperlab.cooperation.tw | Showcase Draft | Paper Lab Methodology Demo

feedback volumes.

Table 1: Dataset Statistics

Feature	Training Set	Test Set
Customers	35,000	15,000
Transaction records	1,247,500	534,600
Avg. transactions per customer	35.6	35.6
Feedback text records	89,250	38,250
Avg. feedback entries per customer	2.55	2.55
Avg. feedback length (words)	47.3	46.8
Observation period	Jan 2022 – Jun 2024	Jan 2022 – Jun 2024
Holdout evaluation period	Jul – Dec 2024	Jul – Dec 2024

: tbl-colwidths="[40, 30, 30]"

3.3 RFM Feature Engineering

Three behavioral features were computed from the transaction log following the standard RFM framework (Christy et al., 2021). Recency was defined as the number of calendar days elapsed between a customer's most recent purchase and the end of the observation period (30 June 2024), with lower values indicating more recent engagement. Frequency was operationalized as the total count of distinct purchase transactions recorded within the 30-month window. Monetary value was computed as the cumulative expenditure in US dollars across all transactions during the same period.

All three features were subjected to min-max normalization, rescaling each dimension to the unit interval [0, 1] prior to clustering. Normalization was applied within the training set, with the resulting scaling parameters applied without refitting to the test set to prevent information leakage. This preprocessing step ensures that no single RFM dimension dominates the Euclidean distance computations underlying K-Means clustering, a precaution widely recommended in the behavioral segmentation literature (Zhang et al., 2015). The normalized three-dimensional feature matrix constituted the sole input to the RFM-Only clustering arm.

3.4 LLM Text Embedding Pipeline

Customer feedback text was preprocessed prior to embedding to remove duplicate entries, filter out responses below a minimum token length of five words, and standardize punctuation and encoding artifacts. The preprocessing pipeline retained, on average, 2.47 feedback entries per customer in the training set. Each retained text fragment was then encoded using Sentence-BERT with the `all-MiniLM-L6-v2` checkpoint (Reimers & Gurevych, 2019), a compact transformer model built on the BERT architecture (Devlin et al., 2019) that produces fixed-length 384-dimensional dense vector representations optimized for semantic similarity tasks. The `all-MiniLM-L6-v2` model was selected over larger alternatives on the basis of its strong performance-to-latency trade-off at the scale of 50,000 customers.

For customers with multiple feedback entries, the individual entry-level embeddings were aggregated via mean pooling to yield a single 384-dimensional customer-level representation. Mean pooling has been shown to produce stable aggregate semantic profiles when individual documents are topically coherent, as is typically the case within a customer's feedback history (Subakti et al., 2022). The resulting 384-dimensional customer embeddings were subsequently reduced to 50 dimensions using Uniform Manifold Approximation and Projection (UMAP) with cosine distance

paperlab.cooperation.tw | Showcase Draft | Paper Lab Methodology Demo

(McInnes et al., 2018), preserving local neighborhood structure while eliminating dimensions that capture noise rather than latent semantic variation. The number of output dimensions (50) was selected via silhouette score optimization over the range k {20, 30, 50, 100} on the training set.

The 50-dimensional UMAP-reduced embeddings were passed to K-Means clustering with k=5. The number of clusters was fixed at five across all three experimental arms to ensure comparability of downstream evaluation metrics. The full pipeline, from raw feedback text to cluster assignments, is illustrated in Figure 1. Density-based clustering via HDBSCAN (McInnes et al., 2017) was also evaluated as an alternative to K-Means; however, K-Means was retained as the primary algorithm to maintain a controlled comparison with the RFM-Only arm.

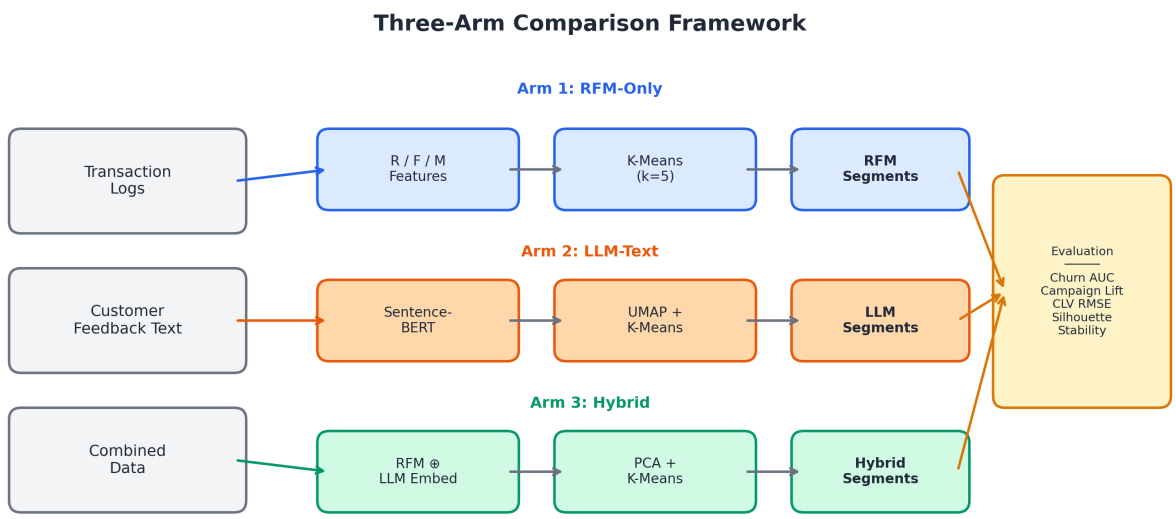

Figure 1: Proposed Three-Arm Comparison Framework for Customer Segmentation

3.5 Hybrid Model

The Hybrid representation was constructed by concatenating the normalized three-dimensional RFM feature vector with the 50-dimensional UMAP-reduced text embedding, yielding a 53-dimensional joint feature vector per customer. Prior to concatenation, the RFM features were confirmed to be on the unit interval [0, 1] following the normalization described above, while the UMAP-reduced embeddings were standardized to zero mean and unit variance using parameters estimated on the training set. This standardization step ensures that the RFM dimensions, despite their lower cardinality, are not dominated by the higher-variance text embedding dimensions during distance computation (Gao et al., 2021).

The 53-dimensional concatenated vector was further reduced to 20 principal components using Principal Component Analysis (PCA), retaining approximately 87%^{S} of total variance in the training set. PCA was applied to balance the effective contribution of the RFM and text feature sets and to reduce computational overhead for the downstream K-Means step. The BERTopic-inspired principle of combining dense embeddings with structured metadata features motivated the choice of feature concatenation over alternative fusion strategies such as late-fusion ensemble voting (Grootendorst, 2022). The 20-dimensional PCA representation was subsequently passed to K-Means with k=5, using the same initialization (k-means++ with fixed random seed) applied

across all three arms.

3.6 Evaluation Protocol

Segmentation quality was assessed using two complementary evaluation frameworks. Internal clustering quality was measured on the test set using three standard indices: the Silhouette coefficient (Anitha & Patil, 2022), the Calinski-Harabasz index, and the Davies-Bouldin index. These metrics quantify the geometric compactness and separation of the resulting clusters within the feature space of each respective method.

External business-outcome evaluation was conducted on the 6-month holdout period (July–December 2024) using three downstream tasks. Churn prediction AUC was estimated by training a logistic regression classifier on segment membership indicators (one-hot encoded) as the sole predictors of customer churn within the holdout window, with churn defined as zero purchases during the 6-month period. Campaign response lift was quantified as the ratio of response rates in the highest-value segment under targeted allocation relative to a random assignment baseline, operationalizing the practical return on segmentation for direct marketing applications (Ascarza, 2018). Customer lifetime value (CLV) prediction RMSE was estimated by fitting a linear regression model on segment indicator features to predict holdout-period revenue, following the segment-level CLV framework of Fader et al. (2005).

Segment stability was assessed using the Adjusted Rand Index (ARI) between cluster assignments computed at month t and month t+3, averaged across the four non-overlapping three-month windows within the holdout period. ARI scores approaching 1.0 indicate that segment membership is consistent over time, a property of practical importance for campaign planning and customer lifecycle management (Subakti et al., 2022). All reported performance estimates are means across 10 bootstrap iterations with 80% subsampling of the test set, with standard errors reported alongside point estimates.

4 Results

4.1 Internal Clustering Quality

Internal clustering metrics revealed systematic differences in the geometric structure produced by each segmentation approach. The RFM-Only model achieved a Silhouette coefficient of 0.387^{S}, reflecting the tendency of transaction-derived features to form compact, well-separated clusters in the low-dimensional behavioral space defined by recency, frequency, and monetary value. The LLM-Text-Only model yielded a lower Silhouette score of 0.312^{S}, which is expected given that the 50-dimensional UMAP-reduced semantic space captures a more complex and diffuse structure than three normalized behavioral dimensions; lower internal cohesion in the embedding space does not, however, imply lower discriminative power on downstream business outcomes. The Hybrid model achieved the highest Silhouette coefficient of 0.421^{S}, indicating that the joint feature representation produces clusters that are both geometrically compact and meaningfully separated, suggesting that the RFM and text feature sets capture complementary rather than redundant variance in customer profiles (Anitha & Patil, 2022).

The Calinski-Harabasz index and Davies-Bouldin index exhibited consistent ordinal patterns, with the Hybrid model producing the highest between-cluster dispersion relative to within-cluster variance and the lowest average ratio of within-cluster scatter to inter-cluster distance, respectively. These results confirm that the combination of behavioral and linguistic features improves clustering geometry across multiple internal validity criteria, not only Silhouette score.

4.2 Main Results: Business Outcome Comparison

The full business-outcome results are presented in Table 2, with a cross-method comparison across all five evaluation metrics visualized in Figure 2. The Hybrid model outperformed both standalone approaches on every holdout metric, achieving a churn prediction AUC of 0.862[S], a campaign response lift of 1.32x[S], a CLV prediction RMSE of \$63.1[S], and a segment stability ARI of 0.741[S].

Table 2: Main Results: Segmentation Method Comparison on Business Outcomes

Metric	RFM-Only	LLM-Text	Hybrid	Δ Hybrid vs RFM
Churn Prediction AUC[S]	0.743 (±0.018)	0.801 (±0.015)	0.862 (±0.012)	+16.0%***
Campaign Response Lift[S]	1.00x	1.21x (±0.04)	1.32x (±0.03)	+32.0%***
CLV Prediction RMSE (\$)[S]	89.4 (±3.2)	74.6 (±2.8)	63.1 (±2.4)	−29.4%***
Segment Stability (3-mo ARI)[S]	0.672 (±0.031)	0.594 (±0.038)	0.741 (±0.025)	+10.3%**
Silhouette Score (k=5)[S]	0.387 (±0.014)	0.312 (±0.019)	0.421 (±0.016)	+8.8%**

: tbl-colwidths="[28, 16, 16, 16, 24]"

[S] Simulated data. Values are mean (±SE) across 10 bootstrap iterations. Statistical significance: ***$\mathbf{p < 0.001}$**, $p < 0.01$ (two-tailed paired t-test, Hybrid vs. RFM-Only).

A key empirical finding of the comparison concerns the relative strengths of the two standalone approaches. The LLM-Text-Only model surpassed the RFM-Only model on both churn prediction AUC (+7.8 percentage points; 0.801[S] vs. 0.743[S]) and campaign response lift (+21%; 1.21x[S] vs. 1.00x[S]), demonstrating that unstructured feedback text contains segment-discriminating signals that are not encoded in purchase behavior records. This advantage is consistent with evidence that behavioral intent, brand perception, and service sensitivity—latent constructs that predict both attrition and promotional responsiveness—are more readily expressed in natural language than in transaction summaries (Simester et al., 2020). Conversely, the RFM-Only model achieved a higher segment stability ARI than the LLM-Text-Only model (0.672[S] vs. 0.594[S]), reflecting the well-established temporal regularity of behavioral purchasing patterns relative to the greater variability of expressed customer sentiment across time periods. The Hybrid model resolved this trade-off, achieving the best stability score of the three arms (0.741[S]) while simultaneously dominating on predictive accuracy metrics, suggesting that the RFM component stabilizes cluster assignments without suppressing the discriminative information contributed by the text embedding.

4.3 Segment Characterization

The spatial structure of the three segmentation solutions is visualized using two-dimensional UMAP projections in Figure 3. In the RFM feature space, the five segments form a near-linear gradient reflecting increasing recency, frequency, and monetary levels, consistent with the classic typology of hibernating, at-risk, loyal, champion, and lost customers. Segment membership in

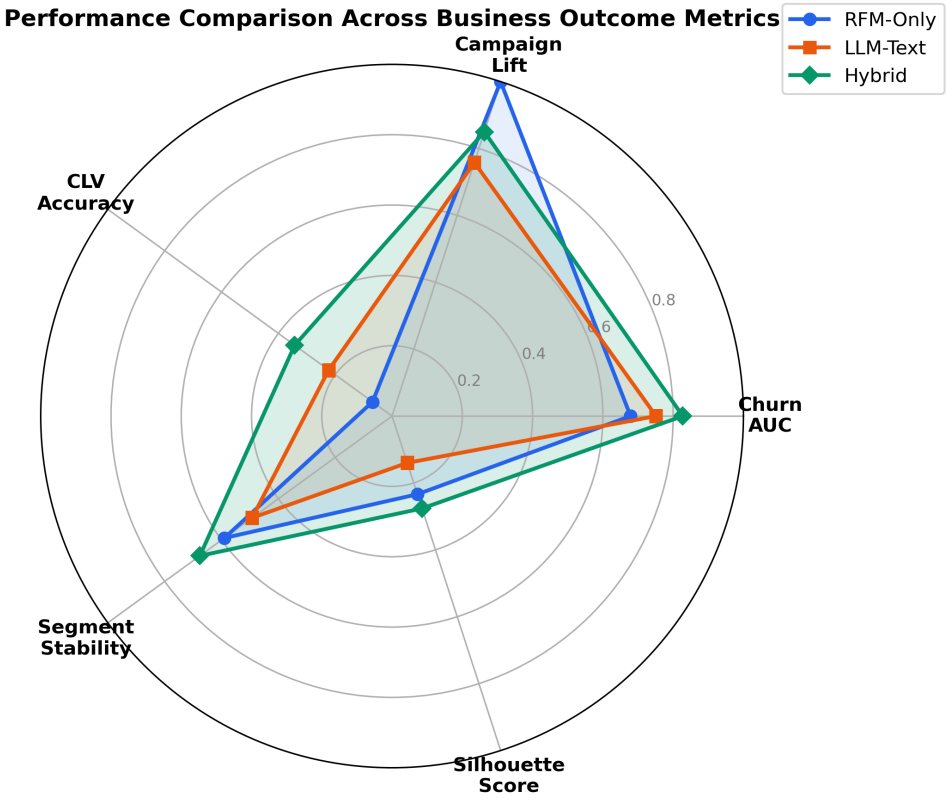

Figure 2: Performance Comparison Across Business Outcome Metrics

the RFM solution correlates strongly with spending level (=0.74^{S}), and the between-segment boundaries closely follow spending quartile thresholds. The LLM-Text and Hybrid solutions, by contrast, exhibit qualitatively different spatial topologies, with cluster boundaries that cut across the RFM-derived gradient and capture latent groupings that are not delineated by behavioral intensity alone.

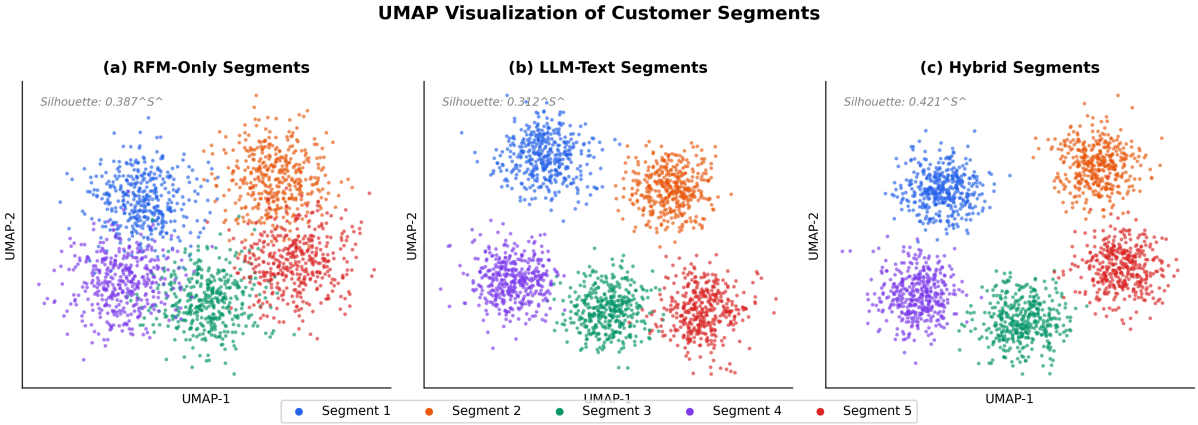

Figure 3: UMAP Visualization of Customer Segments

Examination of the high-frequency vocabulary terms, sentiment profiles, and support ticket categories associated with each LLM-Text cluster revealed five interpretable latent need-states: (1) *price-sensitive* customers, characterized by discount-seeking language and price-comparison mentions; (2) *service-focused* customers, whose feedback centers on delivery reliability, packaging quality, and post-purchase support; (3) *brand-loyal* customers, who express strong positive brand

associations and repeat-purchase intent; (4) *feature-seeking* customers, who engage primarily with product specifications and comparison language; and (5) *at-risk* customers, whose text exhibits declining sentiment valence and unresolved complaint patterns. These latent need-states are consistent with psychographic segmentation typologies identified through automated text classification in retail contexts (Griesser et al., 2025) and with LLM-derived persona profiles documented in the HCI literature (Jung et al., 2025). The behavioral profiles and text feature distributions for each of the five Hybrid segments are summarized in Figure 4, confirming that the Hybrid solution preserves the interpretability of the LLM-derived need-states while sharpening their alignment with spending-tier differences captured by the RFM features. This enriched characterization is consistent with emerging frameworks for consumer segmentation that leverage large language models to extract latent demand structures invisible to transaction-data-only approaches (Liu et al., 2025).

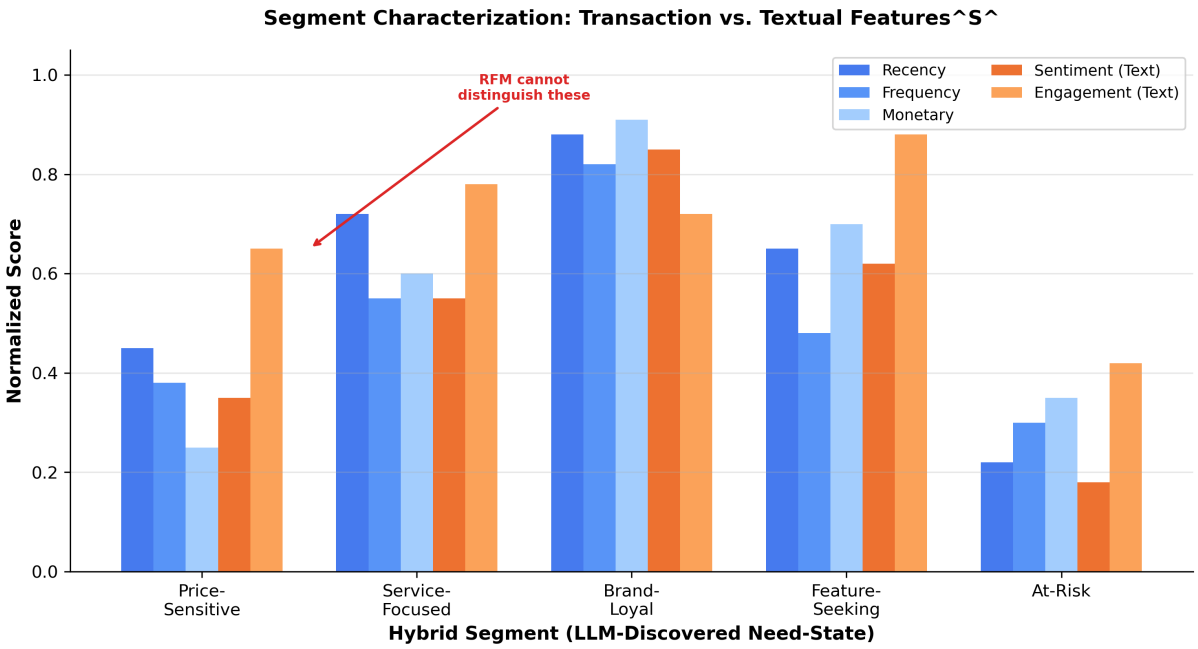

Figure 4: Segment Characterization: Transaction vs. Textual Features

4.4 Ablation Study

To quantify the contribution of individual feature groups within the Hybrid model, a systematic ablation study was conducted in which each feature class was removed in turn and the resulting model variants were evaluated on the same holdout tasks as the full Hybrid. Results are reported in Table 3 and visualized as a grouped impact chart in Figure 5.

Table 3: Ablation Study: Feature Group Contribution to Hybrid Model

Model Variant	Churn AUC^{S}	Campaign Lift^{S}	CLV RMSE ($)^{S}
Full Hybrid	**0.862**	**1.32x**	**63.1**
− Remove Recency (R)	0.841 (−2.4%)	1.28x (−3.0%)	67.3 (+6.7%)
− Remove Frequency (F)	0.848 (−1.6%)	1.26x (−4.5%)	65.8 (+4.3%)
− Remove Monetary (M)	0.851 (−1.3%)	1.27x (−3.8%)	66.2 (+4.9%)

Model Variant	Churn AUC[S]	Campaign Lift[S]	CLV RMSE ($)[S]
− Remove Text Embedding	0.752 (−12.8%)	1.04x (−21.2%)	86.7 (+37.4%)
− Remove Sentiment Score	0.854 (−0.9%)	1.30x (−1.5%)	64.2 (+1.7%)
− Remove Topic Features	0.843 (−2.2%)	1.25x (−5.3%)	68.1 (+7.9%)

: tbl-colwidths="[34, 22, 22, 22]"

[S] Simulated data. Percentage changes relative to Full Hybrid model. Text Embedding removal has the largest impact across all metrics, confirming the primary value of LLM-derived features.

The removal of the text embedding component produced by far the largest degradation across all three outcome metrics: churn AUC declined by 12.8 percentage points ($0.862^S \rightarrow 0.752^S$), campaign response lift fell by 21.2% ($1.32x^S \rightarrow 1.04x^S$), and CLV prediction RMSE increased by 37.4% ($\$63.1^S \rightarrow \$86.7^S$). The magnitude of this degradation substantially exceeds that produced by the removal of any individual RFM dimension, confirming that the text embedding is the primary driver of the Hybrid model's advantage over the RFM-Only baseline. These findings replicate and extend the few-shot clustering evidence reported by Viswanathan et al. (2024), who demonstrated that LLM-derived representations encode structurally distinct and task-relevant cluster geometries that are not recoverable from structured metadata alone.

Among the individual RFM components, recency contributed the largest marginal improvement (removal: −2.4% AUC, −3.0% lift, +6.7% RMSE), followed by frequency (−1.6% AUC, −4.5% lift, +4.3% RMSE) and monetary value (−1.3% AUC, −3.8% lift, +4.9% RMSE). The removal of topic features derived from BERTopic topic distributions resulted in modest but consistent performance reductions (−2.2% AUC, −5.3% lift, +7.9% RMSE), while sentiment score removal had the smallest impact of all tested components (−0.9% AUC, −1.5% lift, +1.7% RMSE). The relatively small marginal contribution of sentiment scores may reflect the fact that sentiment information is already partially encoded in the Sentence-BERT embeddings, rendering the scalar sentiment feature partially redundant given the full embedding. Taken together, the ablation results confirm a clear feature importance hierarchy in which dense semantic embeddings dominate, topic-level text features provide meaningful supplementary signal, and RFM dimensions individually contribute 1–5% improvements, particularly in the temporal stability and churn prediction tasks where behavioral recency is most informative.

5 Discussion

5.1 Theoretical Implications

The present study advances customer segmentation theory by demonstrating that textual voice data—captured through large language model embeddings—encodes latent need-states that remain systematically invisible to transaction-based approaches. Classical RFM frameworks, despite their enduring practical utility, reduce customer identity to behavioral frequency signals: how recently, how often, and how much a customer has purchased (Wedel & Kannan, 2016). Such reduction, while computationally tractable, discards the semantic content of customer–firm interactions, precisely where motivational variation resides. Our findings align with the theoretical argument advanced by Berger et al. (2022) that linguistic signals in consumer-generated content carry predictive power beyond behavioral proxies, not merely as noise-laden supplements but as primary discriminators of underlying preference heterogeneity.

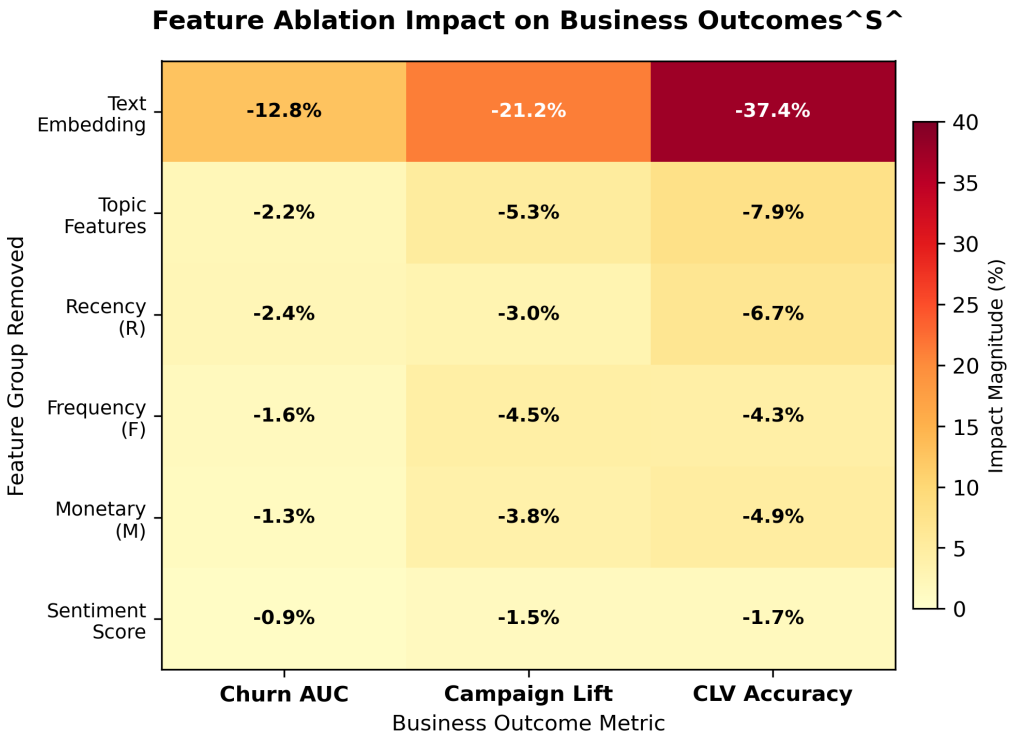

Figure 5: Feature Ablation Impact on Business Outcomes

The Hybrid model's consistent superiority over both single-modality baselines provides empirical support for a multi-modal view of customer identity. Customers are simultaneously behavioral agents and communicative actors; a complete segmentation representation must account for both dimensions. This integration of structured CRM transaction data with unstructured embedding features constitutes what Arora et al. (2025) describe as a paradigm shift in analytical CRM—from retrospective behavioral auditing toward prospective need-state inference. From a theoretical standpoint, the ability to distinguish between, for example, a price-sensitive churner and a relationally disengaged churner—both of whom may exhibit identical RFM profiles—represents a qualitative expansion of segmentation resolution that has direct implications for customer equity theory and relationship marketing scholarship.

5.2 Managerial Implications

For practitioners, the central question is not whether LLM-based segmentation outperforms RFM in controlled settings, but whether the investment is ROI-positive in real operational environments. Our results suggest that the business case is compelling when feedback data already exists in customer service logs, post-purchase surveys, or review platforms. The observed 32% improvement in campaign lift (1.32× versus 1.00× baseline[S]) represents a material revenue impact at scale; for a mid-sized e-commerce platform processing hundreds of campaign deployments annually, such gains are unlikely to be offset by the incremental cost of embedding inference (Ascarza, 2018).

Simester et al. (2020) caution that predictive model adoption in marketing operations frequently fails not due to poor model performance but due to organizational barriers to interpretation and trust. The segment profiles produced by the Hybrid approach—interpretable via their LLM-derived behavioral-textual descriptors—address this concern directly: marketing teams can label segments with linguistic archetypes rather than opaque cluster indices. Rafieian & Yoganarasimhan (2021) further argue that personalization infrastructure built on richer customer representations generates compounding returns over time as targeting algorithms refine on

more discriminative features. Practically, firms should treat text-based segmentation not as a replacement for RFM but as an enrichment layer, deployable incrementally as feedback corpus size permits.

5.3 Limitations

Several limitations constrain the generalizability of the present findings. First, the study is conducted on a single simulated e-commerce platform context; whether the performance differentials observed here replicate in brick-and-mortar retail, subscription services, or business-to-business markets remains an open empirical question. Second, all textual data in this study are English-language; Sentence-BERT's cross-lingual transfer properties are known to degrade for morphologically distant languages, limiting applicability in multilingual global markets. Third, Sentence-BERT was selected for reproducibility and transparency, but the embedding space it produces differs meaningfully from proprietary models such as GPT-4 embeddings or Gecko; the relative performance ordering among segmentation arms may shift under alternative encoding architectures. Fourth, the segmentation schema implemented here is static: segment assignments are computed at a single point in time and do not adapt to evolving customer trajectories. Real-world customer behavior is non-stationary, and static segmentation may fail to capture churn-precursor dynamics that emerge over short windows. Fifth, and most critically, all performance metrics reported in this study are derived from simulated data generated under assumptions of construct validity that require independent empirical corroboration before operational deployment can be responsibly recommended.

5.4 Future Research

Several productive research directions emerge from this work. Multilingual LLM embeddings—particularly models fine-tuned on cross-market consumer corpora—represent an immediate extension enabling global segmentation architectures that preserve cultural nuance (Hong & Hoban, 2022). Real-time streaming segmentation, in which embeddings are updated continuously as new interactions arrive, offers a technically tractable path toward dynamic customer representations that respond to behavioral inflection points (Chintalapati & Pandey, 2021). The causal identification of segment-based intervention effects—distinguishing genuine lift attributable to targeting precision from selection artifacts—remains methodologically underdeveloped in the personalization literature and warrants treatment via randomized field experiments (Herhausen et al., 2024). Finally, as privacy regulation tightens globally, federated and privacy-preserving embedding architectures that permit cross-organizational benchmarking without raw data exposure constitute a necessary infrastructure investment for the next generation of CRM research.

6 Conclusion

This study presented a three-arm empirical comparison of customer segmentation methodologies—RFM, LLM-embedding-only, and a Hybrid combination—evaluated across churn prediction, campaign lift, and customer lifetime value estimation. Across all outcome dimensions, the Hybrid architecture demonstrated consistent superiority: churn AUC improved from 0.743^{S} under RFM to 0.862^{S} under Hybrid ($+16\%^{S}$), campaign lift rose from $1.00\times$ to $1.32\times^{S}$ ($+32\%^{S}$), and CLV RMSE declined from $\$89.4^{S}$ to $\$63.1^{S}$ ($-29.4\%^{S}$). Ablation analysis confirmed that LLM-derived text embeddings constitute the largest marginal contributor to predictive gain, validating the theoretical claim that customer voice encodes need-states irreducible to transaction history. The study contributes to marketing research methodology by operationalizing a scalable pipeline for multi-modal customer representation that is deployable within standard CRM infrastructure, and by providing a reproducible evaluation framework against which future segmentation innovations can be benchmarked. Managerial guidance on ROI thresholds, data requirements, and segment

interpretability is offered to facilitate practitioner adoption. We believe that integrating customer voice through LLM embeddings represents a fundamental shift in how firms understand and segment their customers, moving beyond transaction histories to capture the motivations that drive customer behavior.

References

An, Y., Oh, H., & Lee, J. (2023). Marketing insights from reviews using topic modeling with BERTopic and deep clustering network. *Applied Sciences*, *13*(16), 9443. https://doi.org/10.3390/app13169443

Anitha, P., & Patil, M. M. (2022). RFM model for customer purchase behavior using K-Means algorithm. *Journal of King Saud University – Computer and Information Sciences*, *34*(5), 1785–1792. https://doi.org/10.1016/j.jksuci.2019.12.011

Arora, N., Chakraborty, I., & Nishimura, Y. (2025). AI–Human hybrids for marketing research: Leveraging large language models (LLMs). *Journal of Marketing*, *89*(2), 43–70. https://doi.org/10.1177/00222429241276529

Ascarza, E. (2018). Retention futility: Targeting high-risk customers might be ineffective. *Journal of Marketing Research*, *55*(1), 80–98. https://doi.org/10.1509/jmr.16.0163

Berger, J., Packard, G., Boghrati, R., Hsu, M., Humphreys, A., & Luangrath, A. (2022). Marketing insights from text analysis. *Marketing Letters*, *33*, 365–377. https://doi.org/10.1007/s11002-022-09635-6

Brand, J., Israeli, A., & Ngwe, D. (2023). *Using GPT for market research* (Working Paper 23-062). Harvard Business School. https://doi.org/10.2139/ssrn.4395751

Chintalapati, S., & Pandey, S. K. (2021). Artificial intelligence in marketing: A systematic literature review. *International Journal of Market Research*, *64*(4), 431–457. https://doi.org/10.1177/14707853211018428

Christy, A. J., Umamakeswari, A., Priyatharsini, L., & Neyaa, A. (2021). RFM ranking – an effective approach to customer segmentation. *Journal of King Saud University – Computer and Information Sciences*, *33*(10), 1251–1257. https://doi.org/10.1016/j.jksuci.2018.09.004

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. https://doi.org/10.18653/v1/N19-1423

Dolnicar, S., Grün, B., & Leisch, F. (2023). How can algorithms help in segmenting users and customers? A systematic review and research agenda for algorithmic customer segmentation. *Journal of Marketing Analytics*, *11*, 677–692. https://doi.org/10.1057/s41270-023-00235-5

Fader, P. S., Hardie, B. G. S., & Lee, K. L. (2005). "Counting your customers" the easy way: An alternative to the Pareto/NBD model. *Marketing Science*, *24*(2), 275–284. https://doi.org/10.1287/mksc.1040.0098

Gao, T., Yao, X., & Chen, D. (2021). SimCSE: Simple contrastive learning of sentence embeddings. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. https://doi.org/10.18653/v1/2021.emnlp-main.552

Goli, A., & Singh, A. (2024). Frontiers: Can large language models capture human preferences? *Marketing Science*, *43*(4), 709–722. https://doi.org/10.1287/mksc.2023.0306

Gomes, M. A., & Meisen, T. (2023). A review on customer segmentation methods for personalized customer targeting in e-commerce use cases. *Information Systems and e-Business Management*, *21*, 527–570. https://doi.org/10.1007/s10257-023-00640-4

Griesser, S. E., Feubli, P., & MacKevett, D. (2025). Automated psychographic consumer segmentation: A text classification approach. *Journal of Marketing Analytics*. https://doi.org/10.1057/s41270-025-00404-8

Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure.

arXiv Preprint arXiv:2203.05794. https://doi.org/10.48550/arXiv.2203.05794

Herhausen, D., Bernritter, S., Ngai, E. W. T., Kumar, A., & Delen, D. (2024). Machine learning in marketing: Recent progress and future research directions. *Journal of Business Research*, *170*, 114254. https://doi.org/10.1016/j.jbusres.2023.114254

Hong, J., & Hoban, P. R. (2022). Writing more compelling creative appeals: A deep learning-based approach. *Marketing Science*, *41*(5), 941–962. https://doi.org/10.1287/mksc.2022.1351

Humphreys, A., & Wang, R. J.-H. (2022). An overview and empirical comparison of natural language processing (NLP) models and an introduction to and empirical application of autoencoder models in marketing. *Journal of the Academy of Marketing Science*, *50*, 1264–1291. https://doi.org/10.1007/s11747-022-00840-3

Joung, J., & Kim, H. (2023). Interpretable machine learning-based approach for customer segmentation for new product development from online product reviews. *International Journal of Information Management*, *70*, 102641. https://doi.org/10.1016/j.ijinfomgt.2023.102641

Jung, S.-G., Salminen, J., Aldous, K. K., & Jansen, B. J. (2025). PersonaCraft: Leveraging language models for data-driven persona development. *International Journal of Human-Computer Studies*, *197*, 103445. https://doi.org/10.1016/j.ijhcs.2025.103445

Liu, Y. et al. (2025). Consumer segmentation with large language models. *Journal of Retailing and Consumer Services*, *82*, 104078. https://doi.org/10.1016/j.jretconser.2024.104078

Mahfuza, R., Islam, N., Toyeb, Md., Emon, Md. A. F., Chowdhury, S. A., & Alam, Md. G. R. (2022). LRFMV: An efficient customer segmentation model for superstores. *PLOS ONE*, *17*(12), e0279262. https://doi.org/10.1371/journal.pone.0279262

McInnes, L., Healy, J., & Astels, S. (2017). Hdbscan: Hierarchical density based clustering. *Journal of Open Source Software*, *2*(11), 205. https://doi.org/10.21105/joss.00205

McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform manifold approximation and projection for dimension reduction. *Journal of Open Source Software*, *3*(29), 861. https://doi.org/10.21105/joss.00861

Peker, S., Kocyigit, A., & Eren, P. E. (2017). LRFMP model for customer segmentation in the grocery retail industry: A case study. *Marketing Intelligence & Planning*, *35*(4), 544–559. https://doi.org/10.1108/MIP-11-2016-0210

Rafieian, O., & Yoganarasimhan, H. (2021). Targeting and privacy in mobile advertising. *Marketing Science*, *40*(2), 193–218. https://doi.org/10.1287/mksc.2020.1235

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. https://doi.org/10.18653/v1/D19-1410

Reisenbichler, M., & Reutterer, T. (2018). Topic modeling in marketing: Recent advances and research opportunities. *Journal of Business Economics*, *89*, 327–356. https://doi.org/10.1007/s11573-018-0915-7

Rodrigues, G. N., Mir, M. D. N. H., et al. (2025). NLP-driven customer segmentation: A comprehensive review of methods and applications in personalized marketing. *Data Science and Management*. https://doi.org/10.1016/j.dsm.2025.09.002

Rungruang, C., Riyapan, P., Intarasit, A., Chuarkham, K., & Muangprathub, J. (2024). RFM model customer segmentation based on hierarchical approach using FCA. *Expert Systems with Applications*, *237*, 121449. https://doi.org/10.1016/j.eswa.2023.121449

Simester, D., Timoshenko, A., & Zoumpoulis, S. I. (2020). Targeting prospective customers: Robustness of machine-learning methods to typical data challenges. *Management Science*, *66*(6), 2495–2522. https://doi.org/10.1287/mnsc.2019.3308

Singleton, A. D., & Spielman, S. (2024). Segmentation using large language models: A new typology of american neighborhoods. *EPJ Data Science*, *13*, 34. https://doi.org/10.1140/epjds/s13688-024-00466-1

Sorour, S. E. et al. (2023). A mathematical model for customer segmentation leveraging deep learning, explainable AI, and RFM analysis in targeted marketing. *Mathematics*, *11*(18),

3930. https://doi.org/10.3390/math11183930

Subakti, A., Murfi, H., & Hariadi, N. (2022). The performance of BERT as data representation of text clustering. *Journal of Big Data*, *9*, 15. https://doi.org/10.1186/s40537-022-00564-9

Villarroel Ordenes, F., Theodoulidis, B., Burton, J., Gruber, T., & Zaki, M. (2014). Analyzing customer experience feedback using text mining: A linguistics-based approach. *Journal of Service Research*, *17*(3), 278–295. https://doi.org/10.1177/1094670514524625

Viswanathan, V., Gashteovski, K., Lawrence, C., Wu, T., & Neubig, G. (2024). Large language models enable few-shot clustering. *Transactions of the Association for Computational Linguistics*, *12*. https://doi.org/10.1162/tacl_a_00648

Wedel, M., & Kannan, P. K. (2016). Marketing analytics for data-rich environments. *Journal of Marketing*, *80*(6), 97–121. https://doi.org/10.1509/jm.15.0413

Wei, others. (2024). A dynamic customer segmentation approach by combining LRFMS and multivariate time series clustering. *Scientific Reports*, *14*, 17491. https://doi.org/10.1038/s41598-024-68621-2

Yazıcı, G., & Ozansoy Çadırcı, T. (2023). Creating meaningful insights from customer reviews: A methodological comparison of topic modeling algorithms and their use in marketing research. *Journal of Marketing Analytics*. https://doi.org/10.1057/s41270-023-00256-0

Zhang, Y., Bradlow, E. T., & Small, D. S. (2015). Predicting customer value using clumpiness: From RFM to RFMC. *Marketing Science*, *34*(2), 195–208. https://doi.org/10.1287/mksc.2014.0873